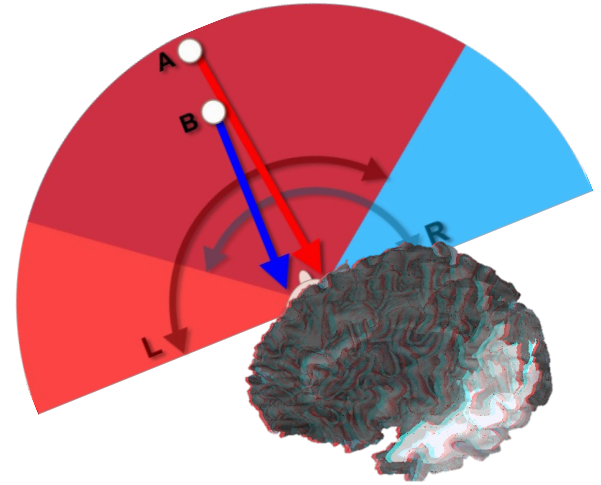


A study of correlation between working memory constraint and declined long-term memory retrieval performance under divided attention



Poopa Kaewbuapan

From last time...

Working memory (WM) is known for having capacity constraint, while long-term memory (LTM) can hold infinitely many items.

The magical number 4 in short-term memory: A reconsideration of mental storage capacity

But while Forsberg and Cowan (2020) [1] concludes that encoding items past a certain set size comes significant LTM retrieval performance degradation later on, questions still arise...

- Load effect alone? What about divide attention?
- How about the retrieval processes side?

Scope and Objectives

- Investigate the effect of both cognitive load and divided attention on the retrieval of long-term memory
 - Design 2 within-subject experimental tasks that assess working memory retrieval performance and long-term memory retrieval performance under 3 conditions that manipulate cognitive load and attentional demand
 - Conduct a 4-days span behavioral data collection of healthy adult population in age range of 18-35
- => Currently finished preliminary behavioral data collection of 2 persons (pilot study)

Day 1 – Working memory retrieval task

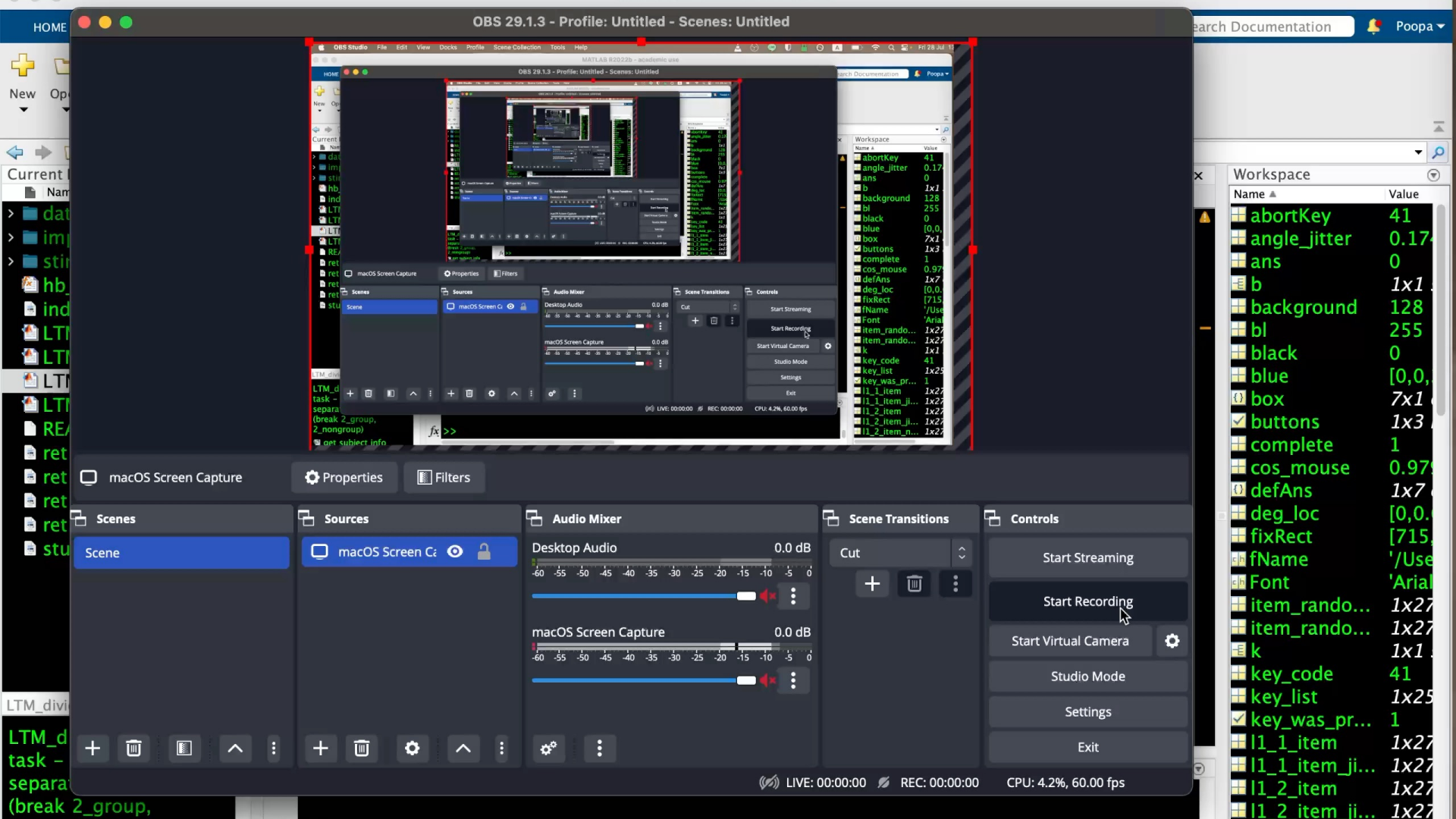
3 conditions

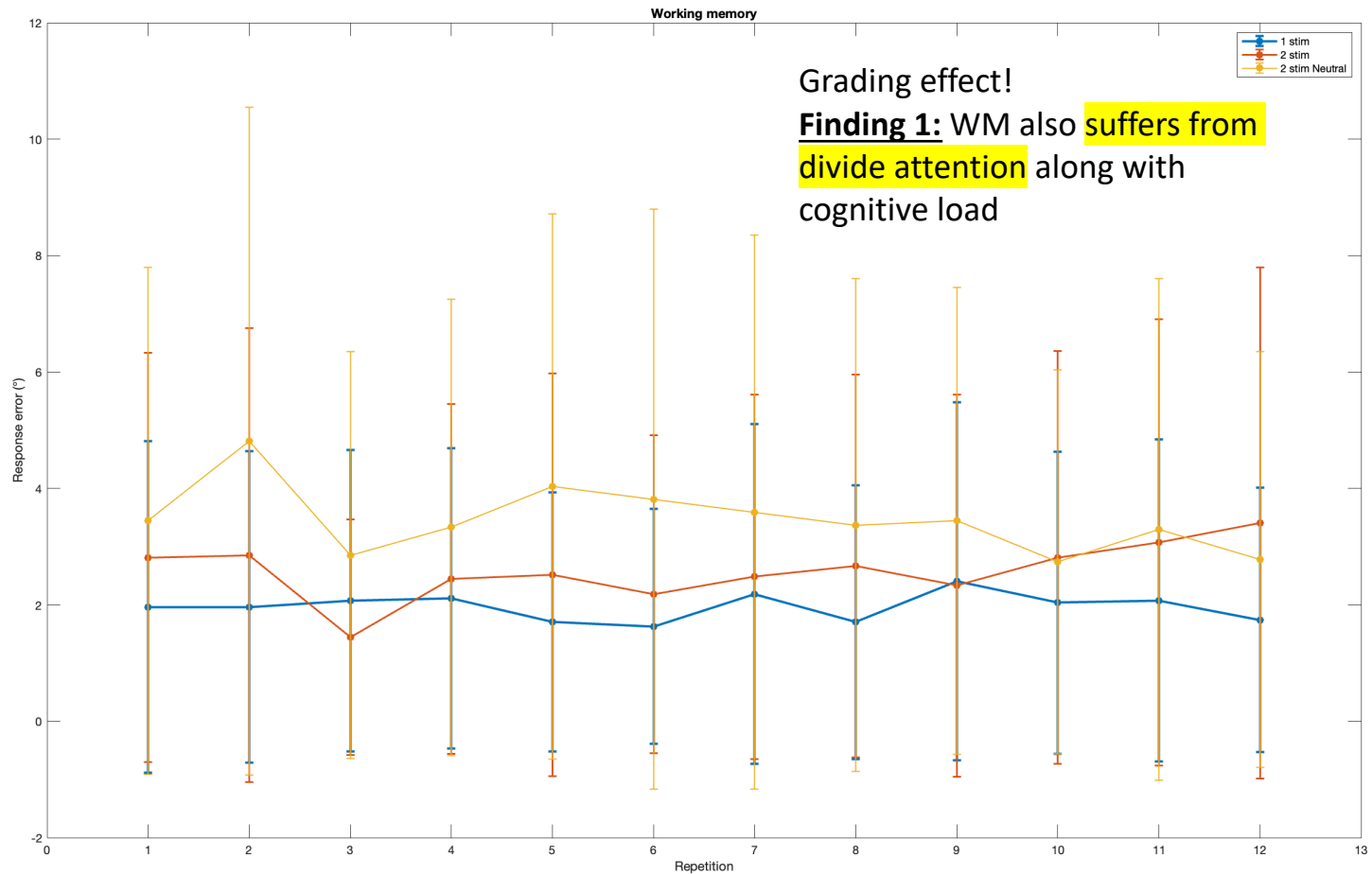
- 1 stimuli => answer **red**, set size 1
- 2 stimuli

=> knowing early if to be asked to answer **red** or **blue**, still set size of 2 to be remember initially [cognitive load manipulation]

- 2 stimuli with distractor - Neutral (purple dot)

⇒ **purple** distractor delayed the response, must remember both locations in WM for a certain time [divide attention manipulation]





Day 2, 3, 4 – LTM memory retrieval task

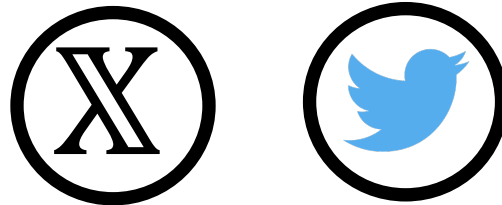
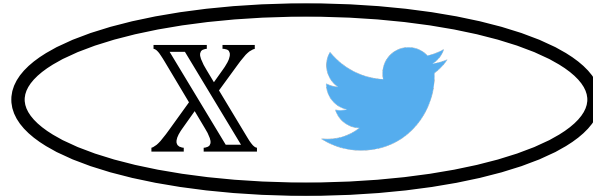
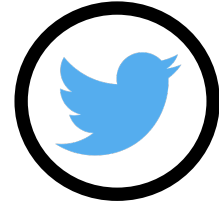
3 conditions {1 recall, 2 Group, 2 Non-Group} with interleaved running design: Study -> Retrieval -> Retrieval individually

- Study => Practice WM recalling of all 162 object-location associations one-by-one (ensures encoding of all spatial associations)
- Retrieval => Given 1 or 2 objects, asked to retrieve the spatial association from LTM. Some trials of 2 objects case will always come in pair (latent property – grouped together)
- Retrieval individually => Separated every 2 objects cue in Retrieval to be presented individually [absence of divide attention]

Day 2, 3, 4 – LTM memory retrieval task

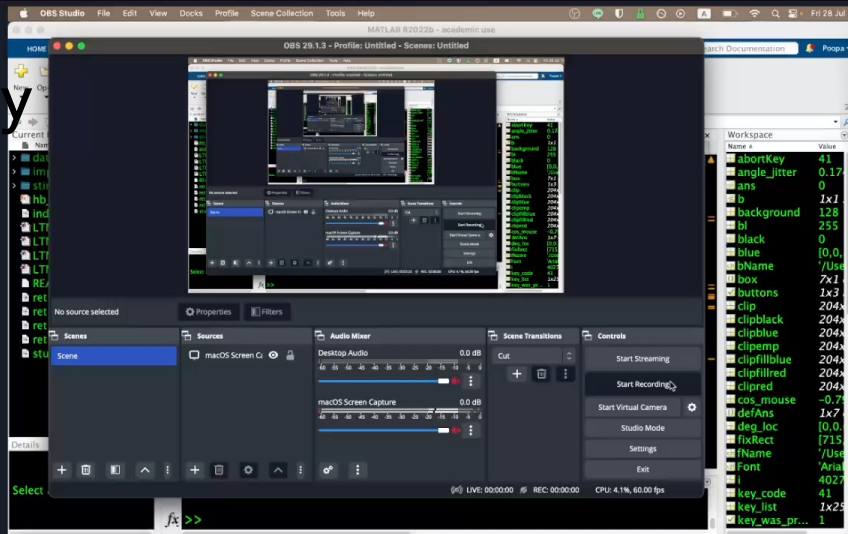
Speculated properties of 3 conditions

- 1 recall => set size 1, single attentional representation
- 2 Group => set size 2 [cognitive load manipulation], single attentional representation
- 2 Non-Group => set size 2, duo attentional representation [divided attention manipulation]



Study

OBS 29.1.3 - Profile: Untitled - Scenes: Untitled



No source selected

Properties

Filters

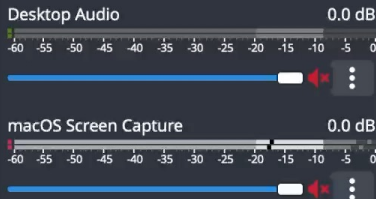
Scenes

Scene

Sources

macOS Screen C

Audio Mixer



Scene Transitions

Cut

Controls

Start Streaming

Start Recording

Start Virtual Camera

Studio Mode

Settings

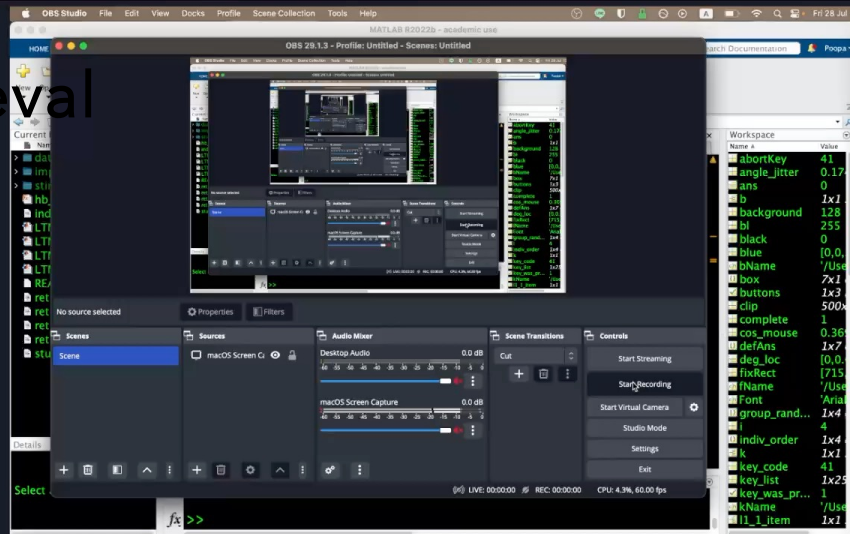
Exit

LIVE: 00:00:00 REC: 00:00:00 CPU: 4.1%, 60.00 fps

Workspace

Name	Value
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angle_jitter	0.17
ans	0
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background	128
bl	255
black	0
blue	[0,0,
bName	"/Use
box	7x1
buttons	1x3
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clipblue	204x
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clipfillred	204x
clipred	204x
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key_list	1x25

Retrieval



No source selected

Properties

Filters

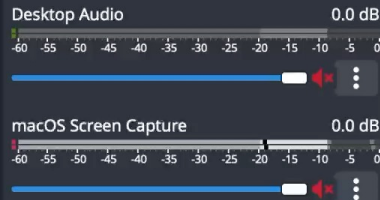
Scenes

Scene

Sources

macOS Screen C

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Scene Transitions

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Controls

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Studio Mode

Settings

Exit

LIVE: 00:00:00 REC: 00:00:00 CPU: 4.3%, 60.00 fps

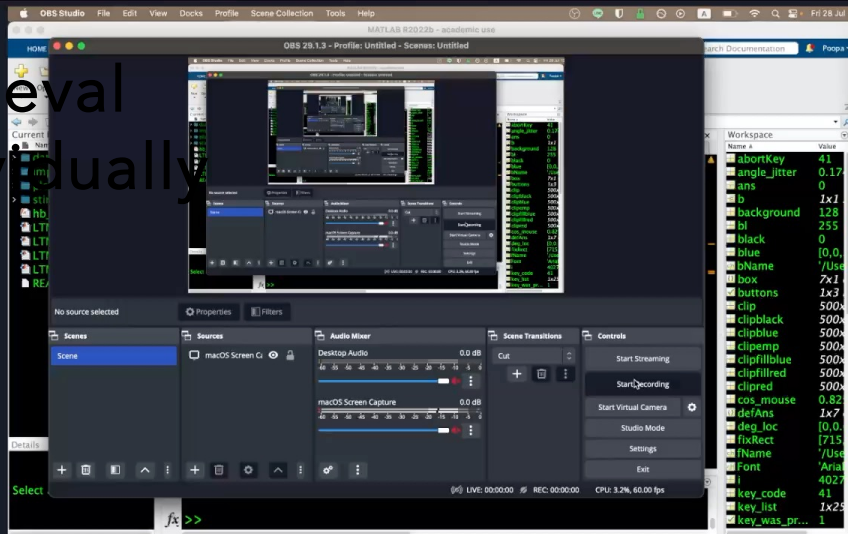
Search Documentation Poopa

Workspace

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background	128
bl	255
black	0
blue	[0,0,
bName	'/Use
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buttons	1x3
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deg_loc	[0,0,
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Font	'Aria
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i	4
indiv_order	1x4
k	1x1
key_code	41
key_list	1x25
key_was_pr...	1
kName	'/Use

Retrieval
individually

OBS 29.1.3 - Profile: Untitled - Scenes: Untitled



No source selected

Properties

Filters

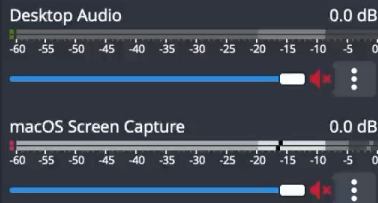
Scenes

Scene

Sources

macOS Screen C...

Audio Mixer



Scene Transitions

Cut

Controls

Start Streaming

Start Recording

Start Virtual Camera

Studio Mode

Settings

Exit

LIVE: 00:00:00 REC: 00:00:00 CPU: 3.2%, 60.00 fps

Search Documentation

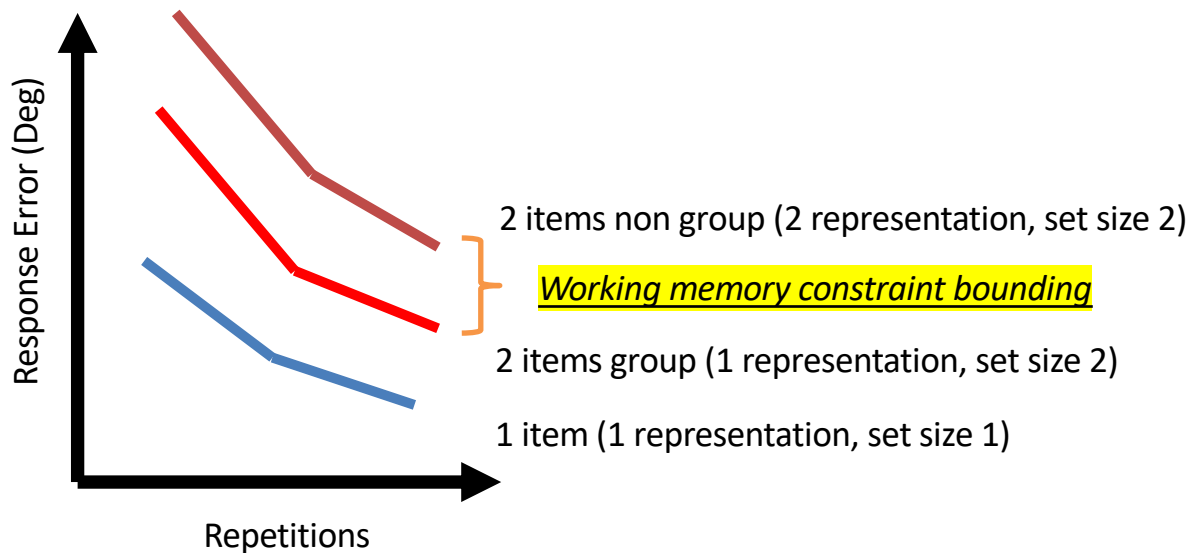
Poopa

Workspace

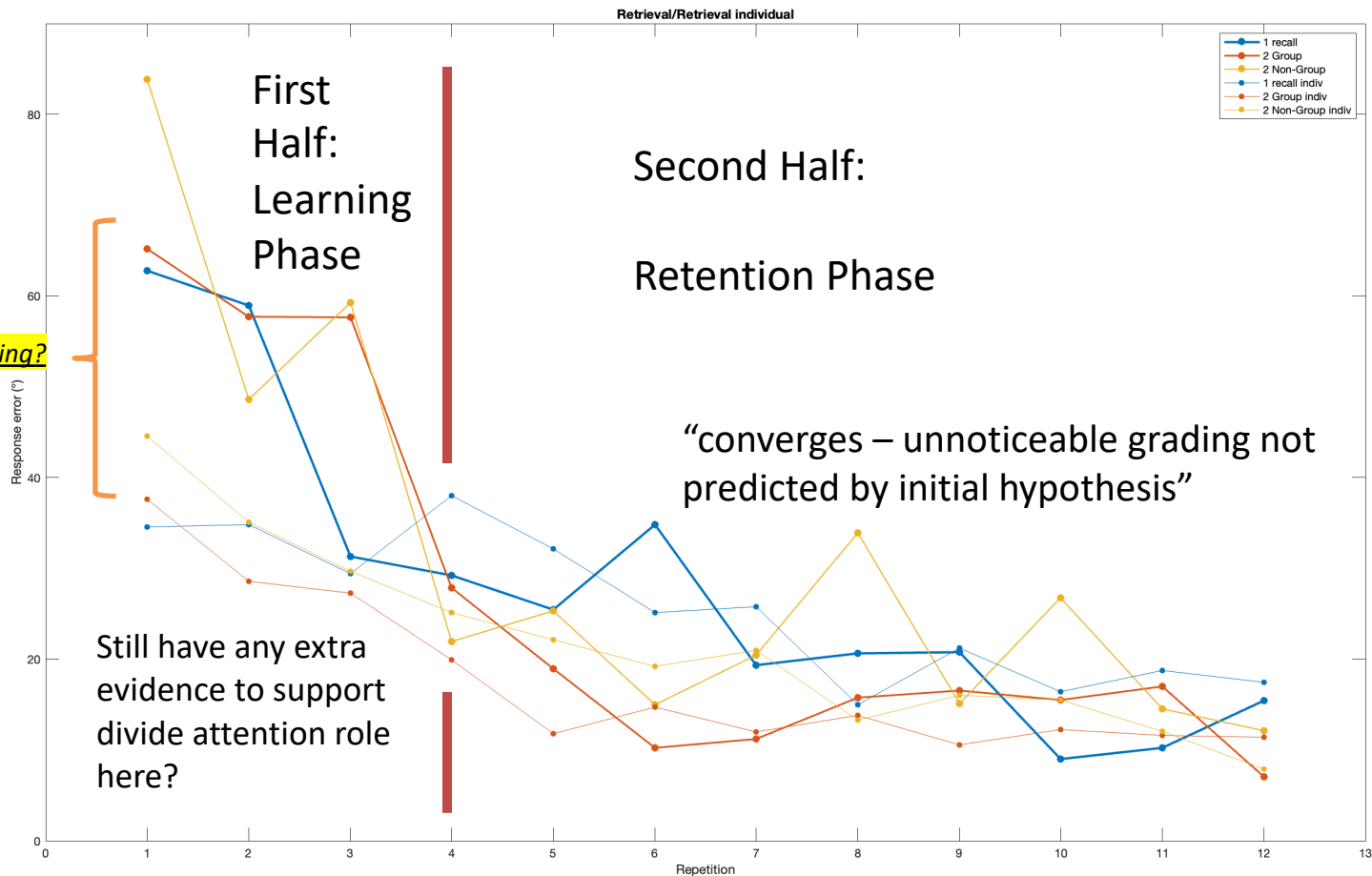
Name	Value
abortKey	41
angle_jitter	0.17
ans	0
b	1x1
background	128
bl	255
black	0
blue	[0,0,
bName	"/Use
box	7x1
buttons	1x3
clip	500x
clipblack	500x
clipblue	500x
clipemp	500x
clipfillblue	500x
clipfillred	500x
clipred	500x
cos_mouse	0.82
defAns	1x7
deg_loc	[0,0,
fixRect	[715,
fName	"/Use
Font	'Aria
i	4027
key_code	41
key_list	1x25

Initial Hypothesis

Working memory constraint impose declination in long-term memory *retrieval* performance under divided attention (set size 2)



WM divide
attention bounding?



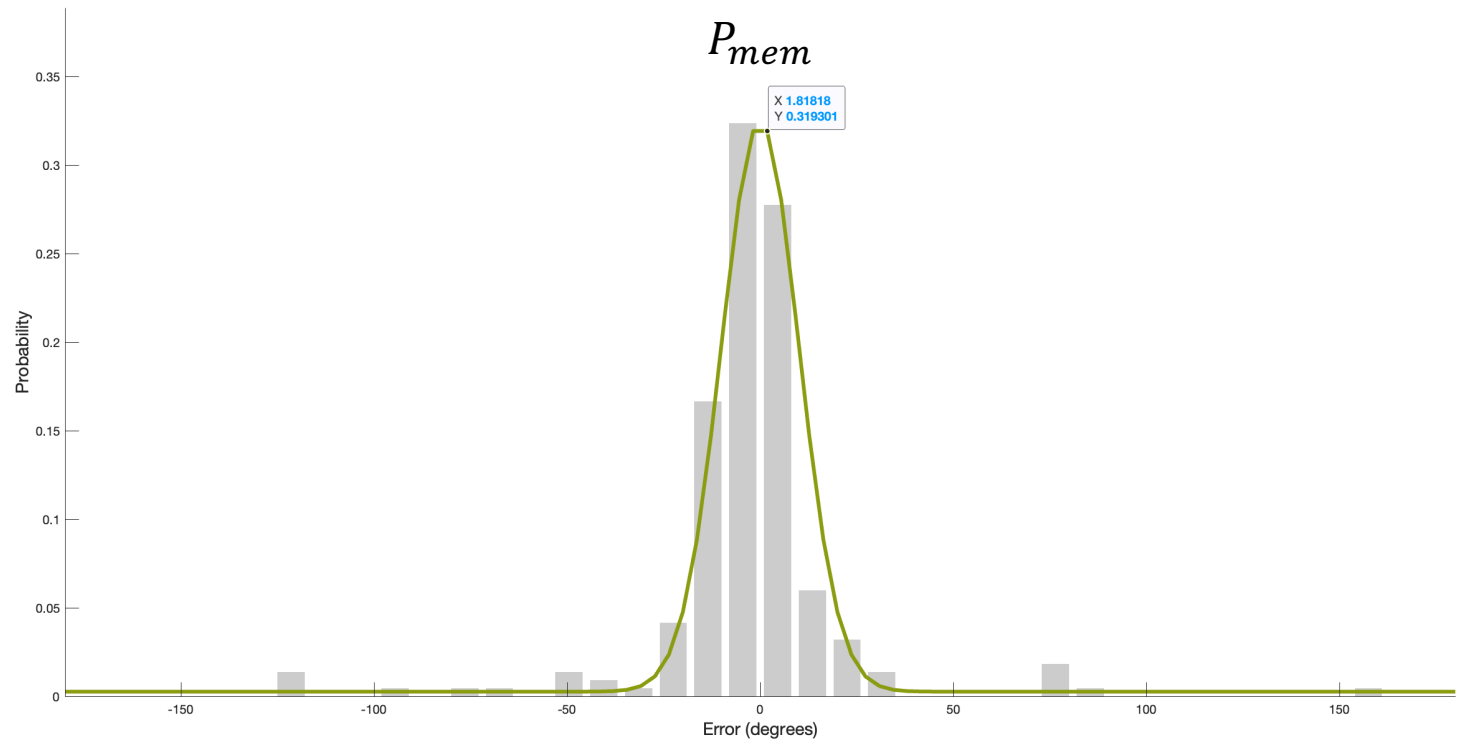
Visual Working Memory Toolbox (MemToolbox)

“The MemToolbox is a collection of MATLAB functions for modeling visual working memory. ...the toolbox includes implementations of popular models of visual working memory, real and simulated data sets, Bayesian and maximum likelihood estimation procedures for fitting models to data ...” [2]

- Reconstructing posterior distribution of probability of retrieval from LTM (Accessibility metric)
- Using Standard Mixture Model which has been used by various past literatures [3]

[2] Suchow, J. W., Brady, T. F., Fougine, D., & Alvarez, G. A. (2013). Modeling visual working memory with the MemToolbox.

[3] D. W. Sutterer, J. J. Foster, J. T. Serences, E. K. Vogel, and E. Awh. (2019). Alpha-band oscillations track the retrieval of precise spatial representations from long-term memory.

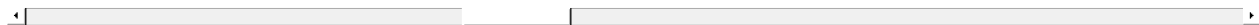


g



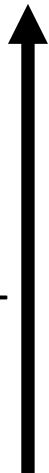
0.11

sd



10.11

Compares with



<u>Retrieval</u>	<u>1st Half</u>	<u>2nd Half</u>
1 recall	0.16	0.236
2 group	0.14	0.32
2 nongroup	0.14	0.25
<u>Retrieval individually</u>		
1 recall	0.16	0.236
2 group	0.18 	0.3
2 nongroup	0.17 	0.3 

*relieved of [cognitive load]

Why?

Probability of retrieval from LTM - Accessibility (more is better)

Second Hypothesis

Speculated properties of 3 conditions might be true

Finding 2: Divided attention might be a possible constraint to WM accessibility and correlates with declined LTM retrieval perf.

At 2nd Half

- Learned Group as already a single attention form and thus retrieval_indiv does not relieve [attentional demand]



- Learned Non-Group as duo attention form, retrieval_indiv task reduces to only single attention form



Retrieval

2 group

2 nongroup

Retrieval individually

2 group

2 nongroup

1st Half

$$P_{mem} = 0.14$$

$$P_{mem} = 0.14$$

$$P_{mem} = 0.18$$

$$P_{mem} = 0.17$$

2nd Half

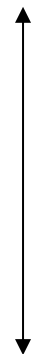
$$P_{mem} = 0.32$$

$$P_{mem} = 0.25$$

$$P_{mem} = 0.3$$

$$P_{mem} = 0.3$$

“Relatively same”



Thank you 🧐